EXISTENCE AND SIX FUNCTORIAL PROPERTIES OF THE DUALIZING COMPLEX OF A PROPER CURVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the relative-dualizing construction with adjointness, the vanishing and CM-support steps, the normalized local criterion, the unwound H^1-duality, and the smooth-case identification are all retained verbatim.

Editorial note. The label lemma-duality-dim-1 names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The relative dualizing complex and its right adjoint. Duality for Schemes, Remark duality-remark-relative-dualizing-complex, Lemma duality-lemma-twisted-inverse-image, Section duality-section-upper-shriek. Defines omega_X^bullet as a(O_Spec(k)) for a the right adjoint of f_*, gives property (4) by construction, and identifies a(O) with f^!(O) for proper f.
- The proper and equidimensional framework examples. Duality for Schemes, Examples duality-example-proper-over-local and duality-example-equidimensional-over-field. Place omega_X^bullet and omega_X in the standard duality framework for a proper morphism to the spectrum of a field.
- Vanishing, CM support, and local structure. Duality for Schemes, Lemmas duality-lemma-vanishing-good-dualizing and duality-lemma-good-dualizing-normalized; Dualizing Complexes, Lemma dualizing-lemma-apply-CM. Yield properties (1), (2) and the cohomologically graded local criterion of (3) through the normalized dualizing complex at closed points.
- The dualizing module and the smooth case. Duality for Schemes, Lemmas duality-lemma-dualizing-module-proper-over-A and duality-lemma-smooth-proper. Give the H^1-duality of property (5) and the identification omega_X = Omega_{X/k} for smooth relative curves in property (6).

Source reference. Original section 0AWJ: sources/stacks-chow/tags.txt, source lines 13722–13722. Prerequisite pointer invoked at proof line(s) 847; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0AWI: sources/stacks-chow/tags.txt, source lines 13721–13721. Prerequisite pointer invoked at proof line(s) 845; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 0AWP: sources/stacks-chow/tags.txt, source lines 13727–13727. Prerequisite pointer invoked at proof line(s) 856; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AWL: sources/stacks-chow/tags.txt, source lines 13724–13724. Prerequisite pointer invoked at proof line(s) 852; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BRT: sources/stacks-chow/tags.txt, source lines 14781–14781. Prerequisite pointer invoked at proof line(s) 860; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AWN: sources/stacks-chow/tags.txt, source lines 13726–13726. Prerequisite pointer invoked at proof line(s) 849; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 0B6S: sources/stacks-chow/tags.txt, source lines 14080–14080. Prerequisite pointer invoked at proof line(s) 837; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0A9Y: sources/stacks-chow/tags.txt, source lines 12966–12966. Prerequisite pointer invoked at proof line(s) 843; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AWS: sources/stacks-chow/tags.txt, source lines 13730–13730. Prerequisite pointer invoked at proof line(s) 854; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a proper scheme of dimension ≤ 1 over a field k. There exists a dualizing complex $\omega_X^\bullet$ with the following properties*

1. *$H^i(\omega_X^\bullet)$ is nonzero only for $i = -1, 0$,*
2. *$\omega_X = H^{-1}(\omega_X^\bullet)$ is a coherent Cohen-Macaulay module whose support is the irreducible components of dimension 1,*
3. *for $x \in X$ closed, the module $H^0(\omega_{X,x}^\bullet)$ is nonzero if and only if either*
   (a) *$\dim(\mathcal{O}_{X,x}) = 0$ or*
   (b) *$\dim(\mathcal{O}_{X,x}) = 1$ and $\mathcal{O}_{X,x}$ is not Cohen-Macaulay,*
4. *for $K \in D_{QCoh}(\mathcal{O}_X)$ there are functorial isomorphisms*[1]
$$\mathrm{Ext}^i_X(K, \omega_X^\bullet) = \mathrm{Hom}_k(H^{-i}(X, K), k)$$
   compatible with shifts and distinguished triangles,
5. *there are functorial isomorphisms $\mathrm{Hom}(\mathcal{F}, \omega_X) = \mathrm{Hom}_k(H^1(X, \mathcal{F}), k)$ for $\mathcal{F}$ quasi-coherent on X,*
6. *if $X \to \mathrm{Spec}(k)$ is smooth of relative dimension 1, then $\omega_X \cong \Omega_{X/k}$.*

Proof. Denote $f : X \to \mathrm{Spec}(k)$ the structure morphism. We start with the relative dualizing complex
$$\omega_X^\bullet = \omega_{X/k}^\bullet = a(\mathcal{O}_{\mathrm{Spec}(k)})$$

[1] This property characterizes $\omega_X^\bullet$ in $D_{QCoh}(\mathcal{O}_X)$ up to unique isomorphism by the Yoneda lemma. Since $\omega_X^\bullet$ is in $D^b_{Coh}(\mathcal{O}_X)$ in fact it suffices to consider $K \in D^b_{Coh}(\mathcal{O}_X)$.

as described in Duality for Schemes, Remark 0B6S. Then property (4) holds by construction as a is the right adjoint for $f_* : D_{QCoh}(\mathcal{O}_X) \to D(\mathcal{O}_{\mathrm{Spec}(k)})$. Since f is proper we have $f^!(\mathcal{O}_{\mathrm{Spec}(k)}) = a(\mathcal{O}_{\mathrm{Spec}(k)})$ by definition, see Duality for Schemes, Section 0A9Y. Hence $\omega_X^\bullet$ and ω_X are as in Duality for Schemes, Example 0AWI and as in Duality for Schemes, Example 0AWJ. Parts (1) and (2) follow from Duality for Schemes, Lemma 0AWN. For a closed point $x \in X$ we see that $\omega_{X,x}^\bullet$ is a normalized dualizing complex over $\mathcal{O}_{X,x}$, see Duality for Schemes, Lemma 0AWL. Assertion (3) then follows from Dualizing Complexes, Lemma 0AWS. Assertion (5) follows from Duality for Schemes, Lemma 0AWP for coherent $\mathcal{F}$ and in general by unwinding (4) for $K = \mathcal{F}[0]$ and $i = -1$. Assertion (6) follows from Duality for Schemes, Lemma 0BRT. □